# Binomial Distribution

What is Binomial Distribution? Binomial distribution is **a common probability distribution that models the probability of obtaining one of two outcomes under a given number of parameters**. It summarizes the number of trials when each trial has the same chance of attaining one specific outcome.

So, imagine that we tossed the two-side coin 1000 times, then we want to know to how many head we got each time, so we plot the number of head we got on x axis and the time we toss the coin on the y axis.

The **binomial distribution** models the number of **successes** in a fixed number of trials, where:

1. Each trial has **only two outcomes** (success / failure)
2. The probability of success **p stays the same**
3. Trials are **independent**
4. The number of trials **n is fixed**

**Think**: counting how many times something "works" out of several tries.
**Formula:**

$$P(X = k) = C(n, k)p^k(1 - p)^{n-k}$$

For a military mission, engineers are creating an expensive system that relies on an important component.

This component is very delicate and fragile has a probability of failure of 0.32. P(an individual component works) = 0.68

They install 6 of these components in parallel, in such a way that they work independently.

If any one of these components works, the system works.

What is the probability exactly 1 of these components works?

Let X represent the number of components that work

P(X = 1) = ?

X has a binomial distribution with n = 6 and p = 0.68

Recall: The system works if at least 1 of the 6 components works. $(p = 0.68)$

What is $P(X \geq 1)$? (The probability the system works)

$P(X \geq 1) = P(X = 1) + P(X = 2) + \ldots + P(X = 6)$

Possible values of $X$:

$$\underset{X=0}{0}, \underbrace{1, 2, 3, 4, 5, 6}_{X \geq 1}$$

$$P(X \geq 1) = 1 - P(X = 0)$$